



BLOCK URI-205

Lecture (II)



Ass. Prof. Dr. Safaa Said

Department of Histology & Cell
Biology

Assuit University



Learning objectives

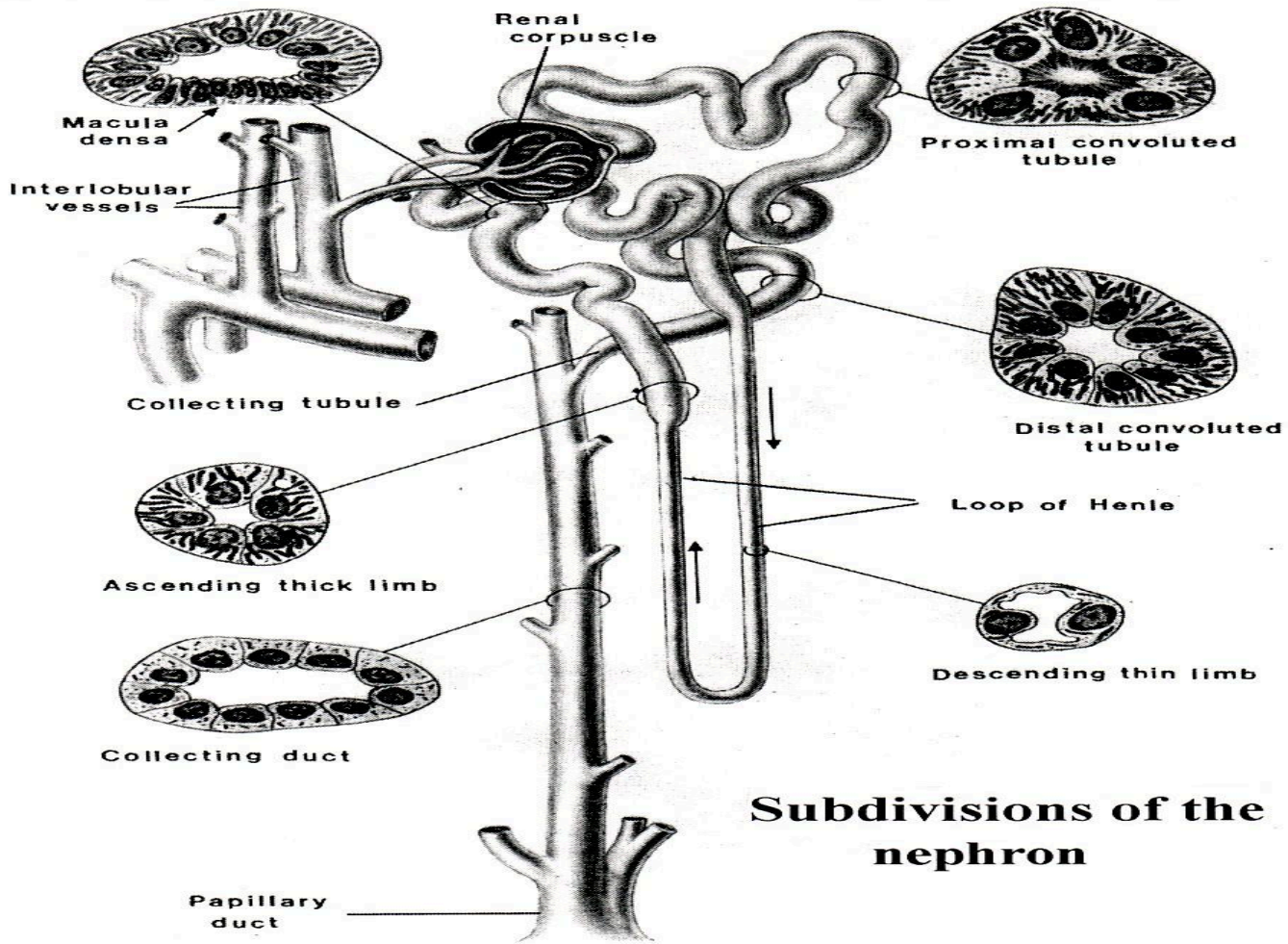
After this lecture, students should be able to:

Describe the histological structure of the proximal convoluted tubules. .

- Compare between the histological structure of the proximal convoluted tubules & distal convoluted tubules . .

- Discriminate loop of Henle , collecting tubules and collecting ducts (the type of their lining epithelium and identify their functions).

- Describe the histological structure of the Juxtaglomerular apparatus and identify its functions.
- Describe the histological structure of the renal interstitium.
- Describe the blood supply of the kidney . .



**Subdivisions of the
nephron**

The renal tubular system

- *is divided into several functional units: proximal convoluted tubule (PCT), loop of Henle, distal convoluted tubule (DCT), collecting tubules, and collecting ducts.*
- **Proximal convoluted tubules** Start at the tubular pole of the renal corpuscle where, the simple squamous epithelium of the capsule's parietal layer is continuous with the simple cuboidal

. *With L.M:*

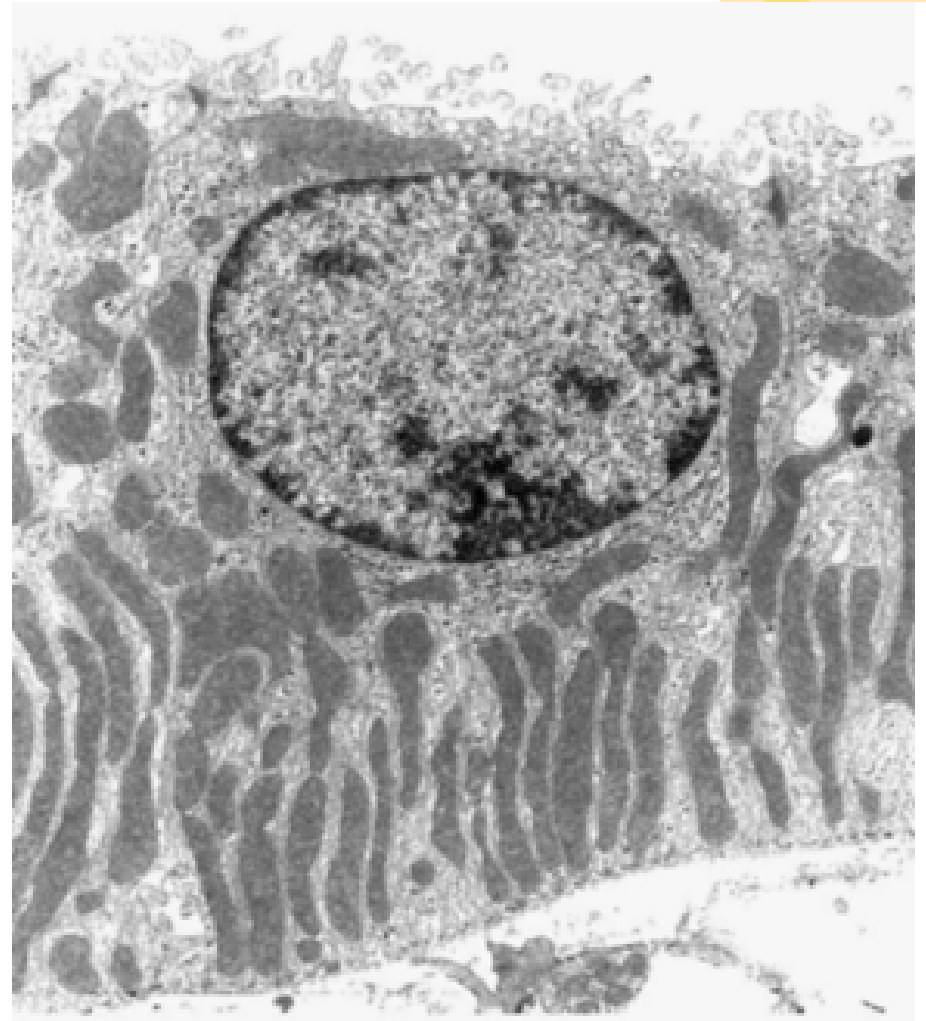
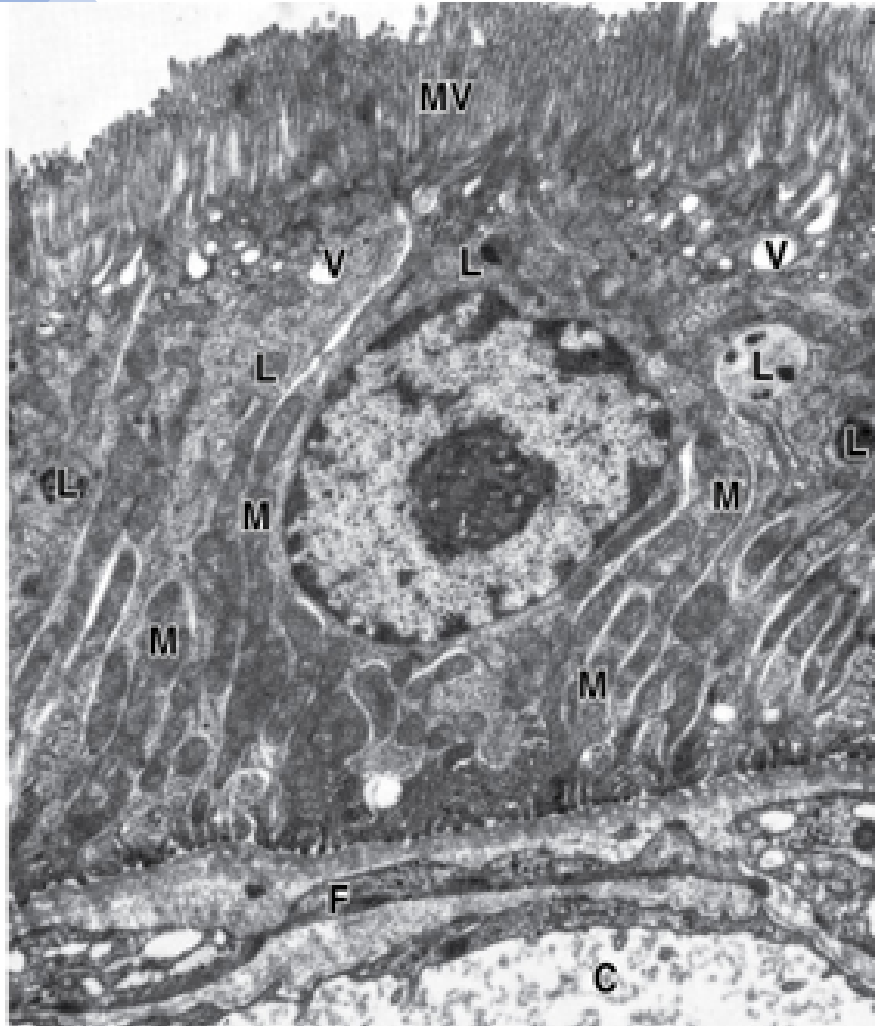
- . The P.C.Ts. are lined with simple cuboidal epithelium.
- . They have brush borders. In routine histologic preparations, the long brush border may be disorganized and give the lumens a fuzz-filled appearance.
- . The cytoplasm is granular and deeply acidophilic.
- . The basal part of the cytoplasm show vertical acidophilic striations.
- . The nuclei are large and spheroidal.
- . Only 3-5 nuclei appear in a cross section.
- . The cell borders are ill- defined.
- . The lumen of the tubules varies according to the activity of the cells.

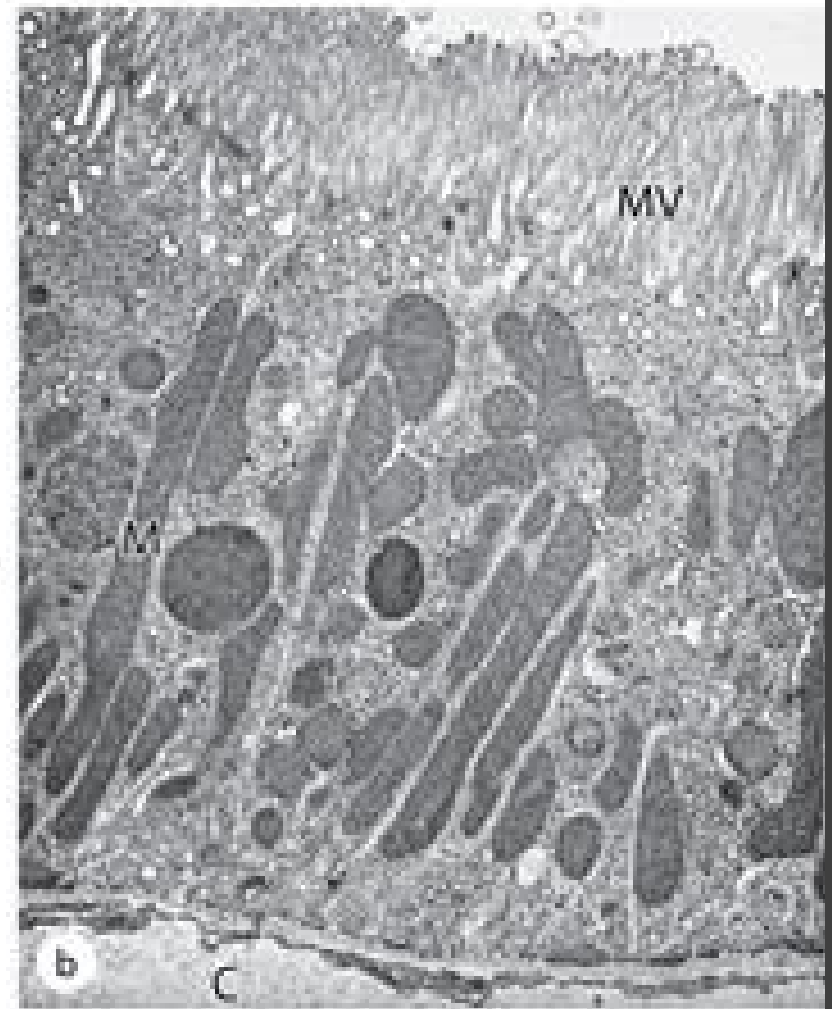
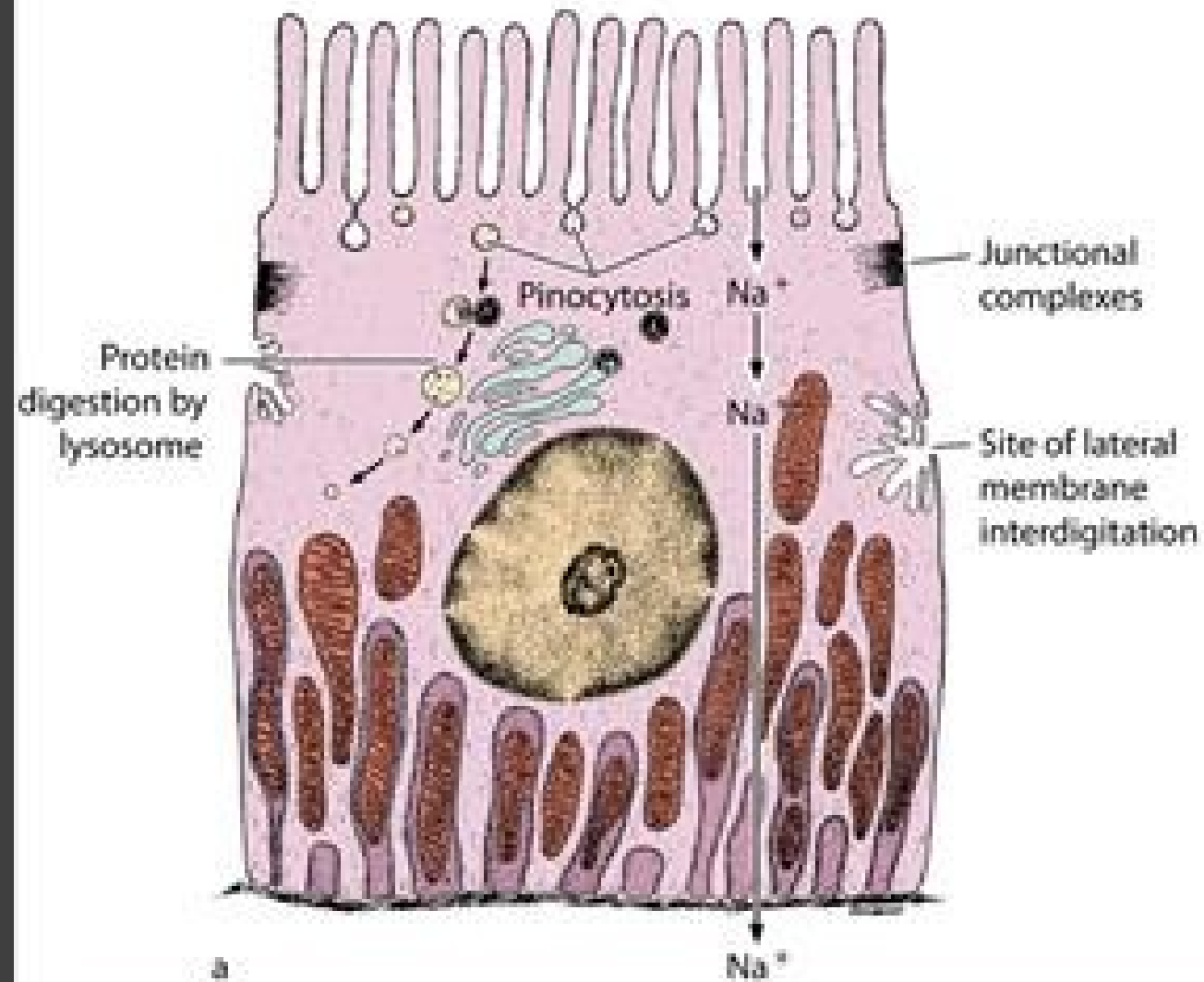
- *E.M. of P.C.Ts.*

- . The luminal border consists of countless microvilli
- . Immediately below the microvilli are many pinocytotic vesicles, indicating active endocytosis and pinocytosis .
- . Interdigitations with neighboring cells are also present laterally.
- . Typically, long invaginations of the basal cell membrane outline regions with mitochondria supply ATP locally for the membrane proteins involved in active transport.
- . Immediately below the basal lamina, blood capillaries remove water absorbed across the epithelium.

- *Functions :*

- The cells of the proximal tubules have major roles in reabsorption and secretion owing to a large variety of transmembrane ion pumps, ion channels, transporters, enzymes, and carrier proteins .
- Cells of the proximal tubule also perform hydroxylation of vitamin D and release to the capillaries.





The Henle's loop

It has four segments:

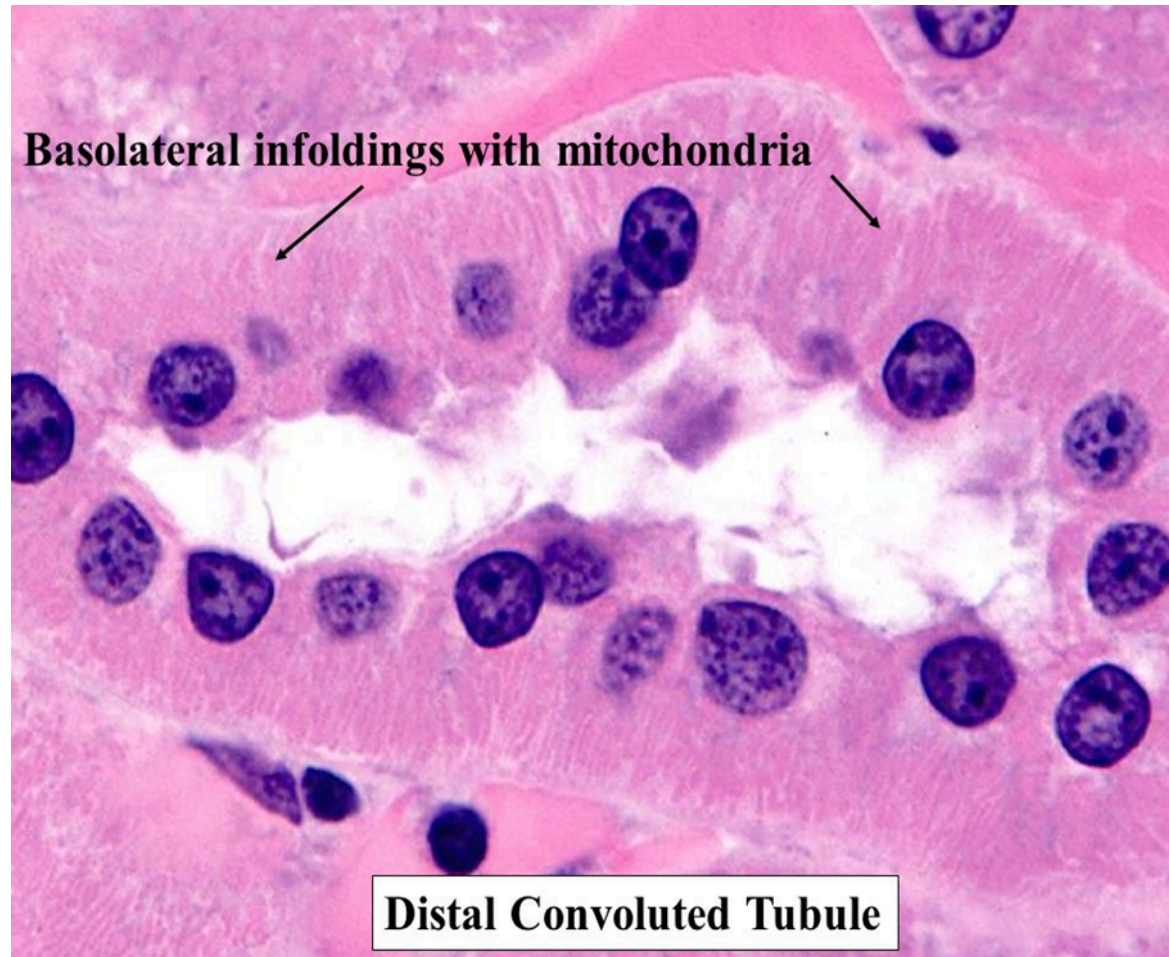
1- **The thick descending limb:** it is a straight continuation of the PCT in the medulla, with a lining epithelium similar to it.

2&3- **The thin descending and the thin ascending limbs:** take a U-shaped course in the medulla, lined by simple squamous epithelium with few organelles.

4- **The thick ascending limb:** it is the straight part of the DCT with its same lining epithelium.

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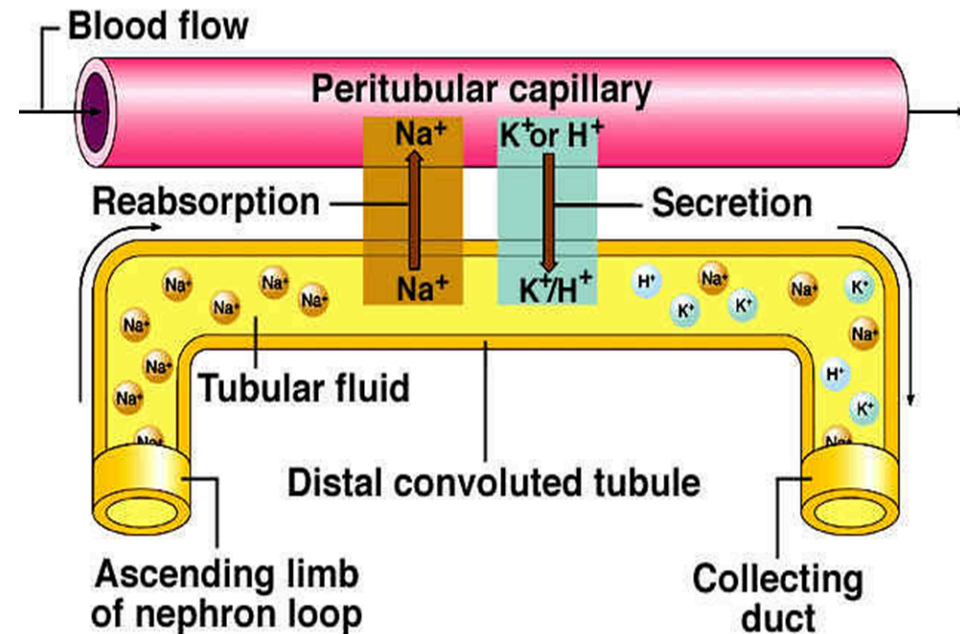
The distal convoluted tubule (DCT)

- The simple cuboidal cells of the distal tubules differ from those of the proximal tubules in being smaller and having no brush border and more empty lumens.

- **Function:**

1- Facultative reabsorption of water and Na (aldosterone-dependent), secretion of K.

2- Shares in acidification of urine (reabsorption of HCO_3^- & secretion of H^+).



Proximal convoluted tubules P.C.T.	Distal convoluted tubules D.C.T.
<u>L.M.</u> – The P.C.T. is lined with simple cuboidal epithelium. – They have brush borders. In routine histologic preparations, the long brush border may be disorganized and give the lumen a fuzz-filled appearance. – The cytoplasm is granular and deeply acidophilic. – The basal part of the cytoplasm shows vertical acidophilic striation. – The nuclei are large and spheroidal. – Only 3–5 nuclei appear in a cross section. – The cell borders are ill-defined. – The lumen of the tubule varies according to the activity of the cells.	<u>L.M.</u> – The D.C.T. is lined by cuboidal cells which are smaller, more flat than those of PCT. – No brush border. – Less granular cytoplasm. – Less acidophilic cytoplasm and less basal striations. – The number of the cells is (5–8). – The cell borders are well-defined – The lumen is wider.
<u>E.M.</u> – The luminal border consists of countless microvilli, the apical cytoplasm of these cells has numerous pits and vesicles near the bases of the microvilli, indicating active endocytosis and pinocytosis. – Have many long basal membrane invaginations and lateral interdigitations with neighboring cells. – Both the brush border and the basolateral folds contain the many types of transmembrane proteins that mediate tubular reabsorption and secretion. – A large number of giant elongated mitochondria arranged parallel to the long axis of the cells and between the basal infoldings supply ATP locally for the membrane proteins involved in active transport. – Golgi apparatus is located in the supranuclear part of the cytoplasm.	<u>E.M.</u> – Short microvilli. – Less interdigitations. – Less extensive basal infoldings. – Much less tubular reabsorption occurs here than in the proximal tubule. The distal tubule is the primary site of urine concentration.

Proximal convoluted tubules

P.C.T.

LM:

- The P.C.T. is lined with simple cuboidal epithelium.
- They have brush borders. In routine histologic preparations, the long brush border may be disorganized and give the lumens a fuzz-filled appearance.
- The cytoplasm is granular and deeply acidophilic.
- The basal part of the cytoplasm show vertical acidophilic striation.
- The nuclei are large and spheroidal.
- Only 3-5 nuclei appear in a cross section. - The cell borders are ill- defined.
- The lumen of the tubule varies according to the activity of the cells.

Distal convoluted tubules D.C.T.

LM:

- The D.C.T. is lined by cuboidal cells which are smaller, more flat than those of PCT.
- No brush border.
- Less granular cytoplasm.
- Less acidophilic cytoplasm and less basal striations.
- The number of the cells is (5-8). - The cell borders are well- defined - The lumen is wider.

E.M.

- The luminal border consists of countless microvilli, the apical cytoplasm of these cells has numerous pits and vesicles near the bases of the microvilli, indicating active endocytosis and pinocytosis .
- Have many long basal membrane invaginations and lateral interdigitations with neighboring cells .
- Both the brush border and the basolateral folds contain the many types of transmembrane proteins that mediate tubular reabsorption and secretion.
- A large number of giant elongated mitochondria arranged parallel to the long axis of the cells and between the basal infoldings supply ATP locally for the membrane proteins involved in active transport.
- Golgi apparatus is located in the supranuclear part of the cytoplasm.

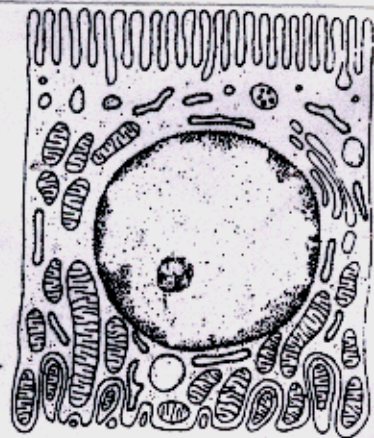
E.M.

- Short microvilli.
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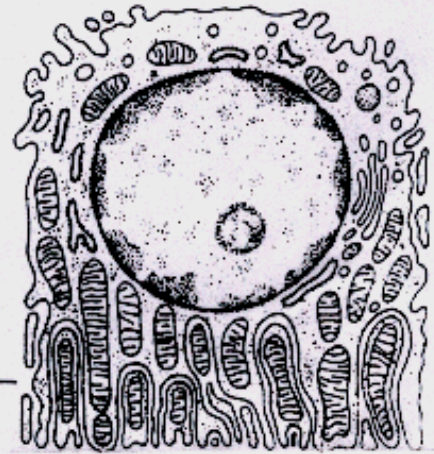
Distal convoluted tubules (DCT)	Proximal convoluted tubules (PCT)
L.M: - The D.C.T. is lined by cuboidal cells which are shorter. - No brush border. - Less granular cytoplasm. - Less acidophilic cytoplasm and less basal striations. - Rounded and slightly apical nuclei. - The number of the cells is larger (5-8). - The cell borders are more distinguishable. - The lumen is wider.	L.M: - The P.C.T. is lined with simple cuboidal epithelium. - They have brush borders. - The cytoplasm is granular and deeply acidophilic. - The basal part of the cytoplasm show vertical acidophilic striation. - The nuclei are large and spheroidal. - Only 3-5 nuclei appear in a cross section. - The cell border is ill defined. - The lumen of the tubule is narrower.

Distal convoluted tubules (DCT)	Proximal convoluted tubules (PCT)
E.M. - Few microvilli. - Less interdigitations. - Less extensive basal infoldings.	E/M. - The luminal border consists of countless microvilli. - The cell membranes of the lateral sides of adjacent cells interdigitate with one another. - The basal part of the cell membrane gives many infolding. -A large number of gaint elongated mitochondria arranged parallel to the long axis of the cells and between the basal infoldings.

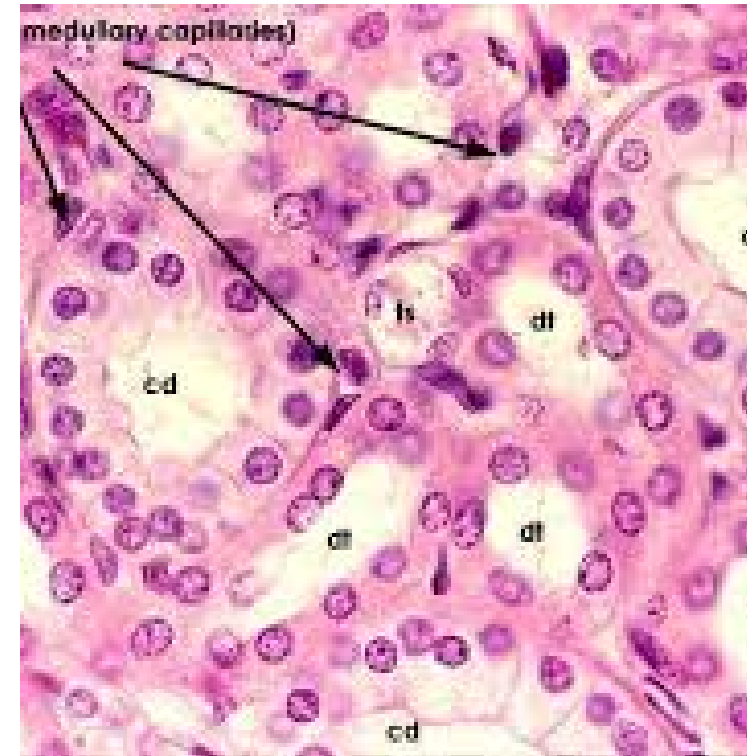
Proximal convoluted
tubule and thick
descending limb of
Henle's loop

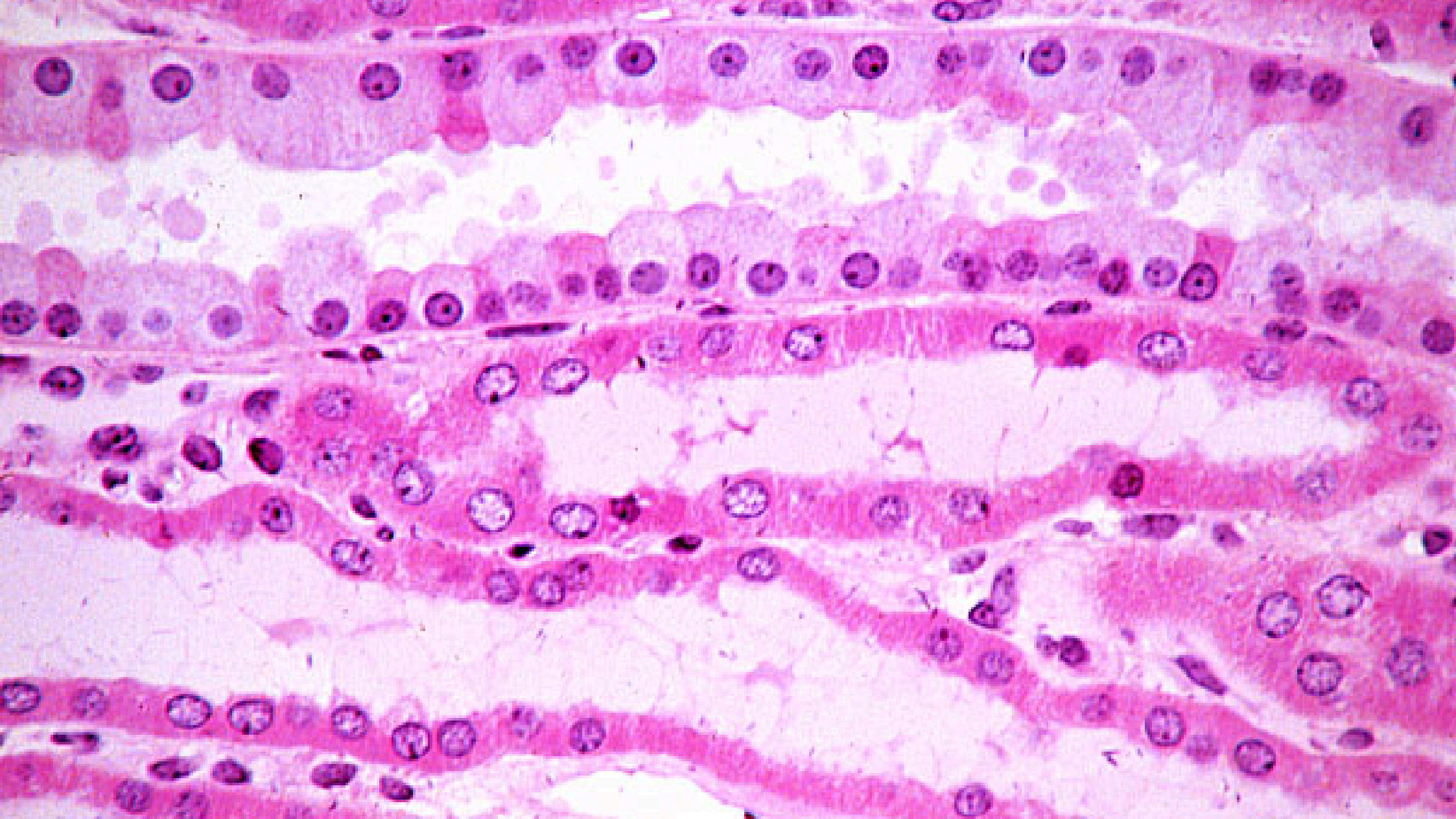


Distal convoluted tubule
and thick ascending limb
of Henle's loop



- ***Collecting tubules and ducts*** are composed mainly of pale staining *principal cells* and *intercalated cells*.
- With L.M. *principal cells* show few organelles, sparse microvilli, and unusually distinct cell boundaries.
- With E.M., these cells are seen to have basal membrane infoldings, consistent with their role in ion transport. Principal cells are particularly rich in aquaporins, the integral membrane pore proteins that function as specific channels for water molecules.

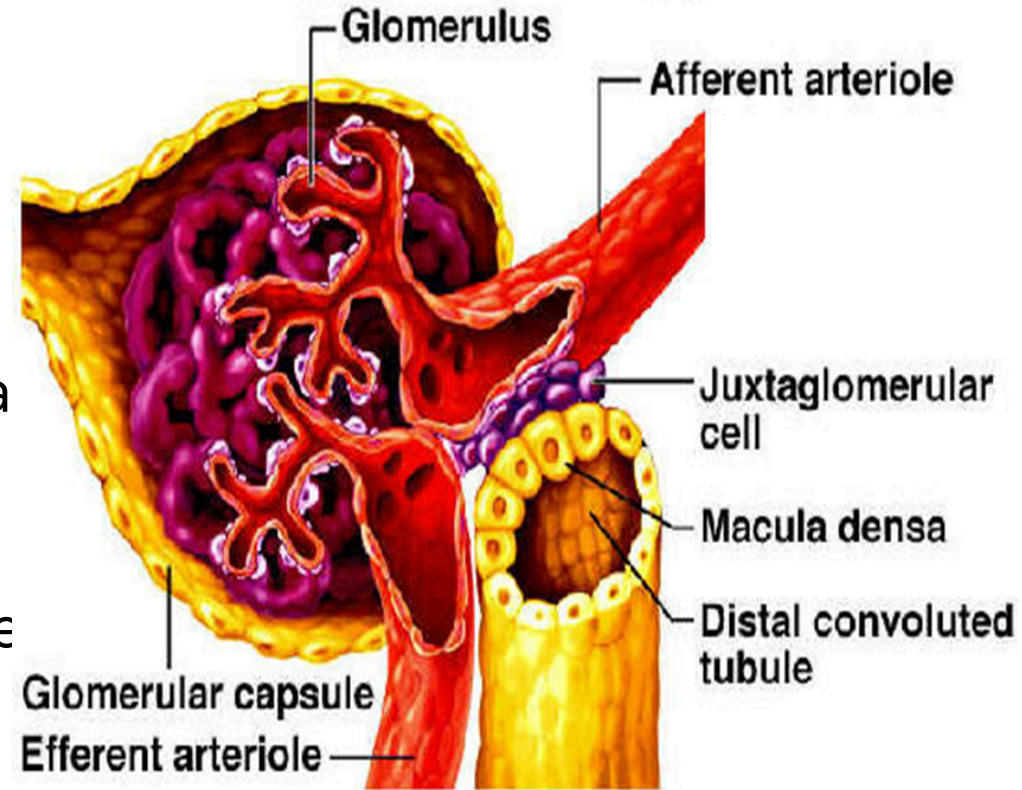




- A *connecting tubule* extends from each nephron and several join together in the cortical medullary rays to form collecting ducts of simple cuboidal epithelium and an average diameter of 40 μm .
- -In the medulla these merge further, forming larger and straighter collecting ducts with increasingly columnar cells and overall diameters reaching 200 μm . The medullary collecting ducts are the final site of water reabsorption from the filtrate.
- The last part of each nephron, *the connecting tubule*, carries the filtrate into a collecting system that transports it to a minor calyx and in which more water is reabsorbed if needed by the body

Juxtaglomerular apparatus:

- Where the initial, straight part of the distal tubule contacts the arterioles at the vascular pole of the renal corpuscle of its parent nephron, its cells become more columnar and closely packed, forming the macula densa (L. thicker spot).
- This is a part of a specialized sensory structure, the juxtaglomerular apparatus (JGA) that utilizes feedback mechanisms to regulate glomerular blood flow and keep the rate of glomerular filtration relatively constant.



Juxtaglomerular apparatus

- It is formed of ;

• *1-Modification in the afferent arteriole:*

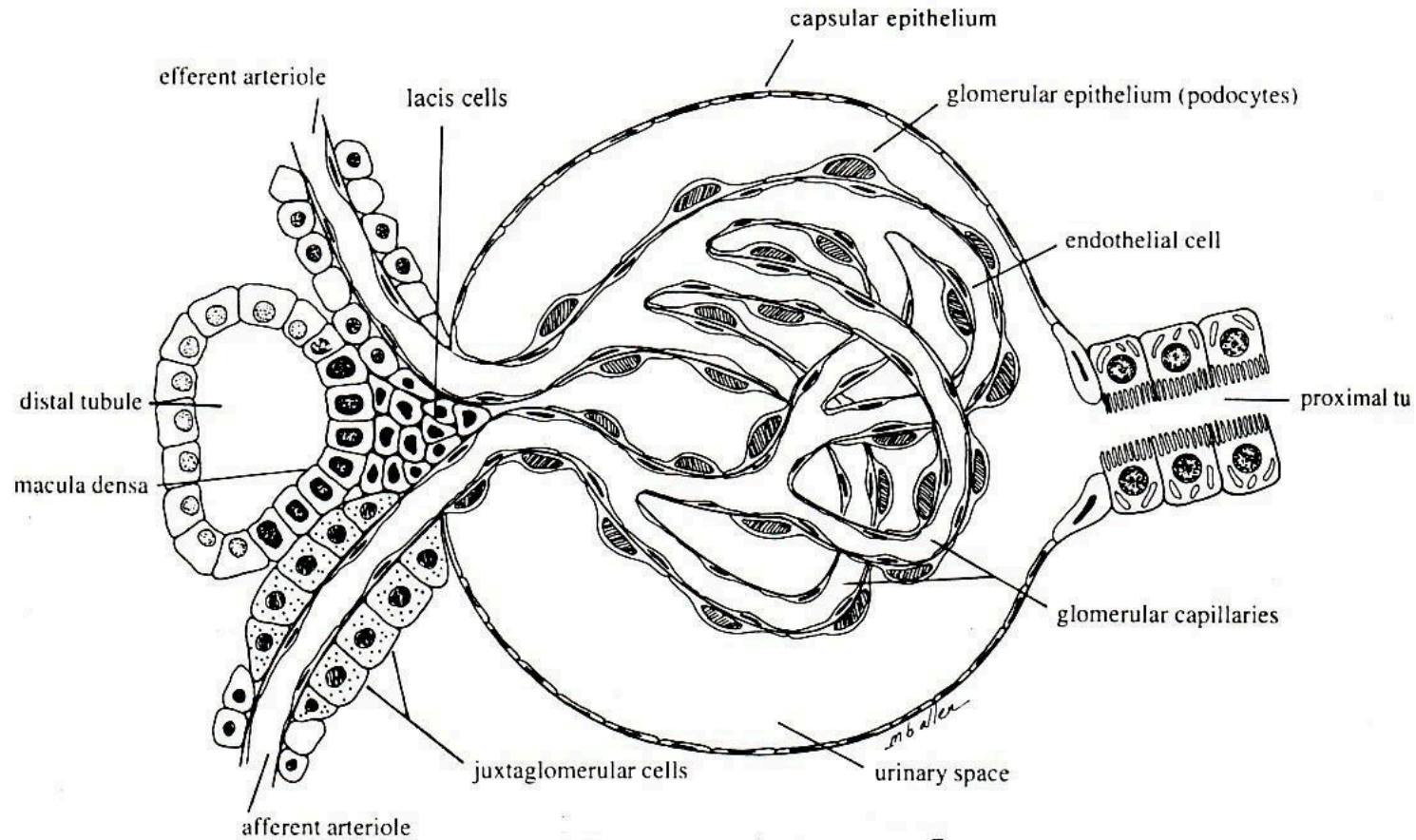
- The smooth muscle cells are modified as **juxtaglomerular granular (JG) cells**, with a secretory phenotype(specialized myoepithelial cells) including more rounded nuclei, rough ER, Golgi complexes, and granules with the protease renin.
- **Function:**
- 1- Secretion renin.
- 2- Secretion of erythropoietin.

• *2-Modification in the distal convoluted tubule:*

- This modification occurs in the portion of the distal convoluted tubule which lies in contact with renal corpuscle.
 - -The basement membrane is lost.
 - -The cells become taller columnar and crowded.
 - -The nuclei become deeply stained and close together.
- These cells are called **macula densa cells**.

3-Extra glomerular mesangial cells.

- Occupy the area between the macula densa, the glomerulus and the afferent arteriole.
- **Function:** Support and coordination (intercellular signaling)
- Basic functions of the JGA is the autoregulation of the GFR and controlling blood pressure via several activities including;
 - **1-Local baroreceptors** in the afferent arteriole, possibly the JG cells themselves and release of renin the initial enzyme in the renin-angiotensinaldosterone system.
 - **2-Monitoring of luminal concentrations** of Na^+ and Cl^- in of the nephron by cells of the macula densa.



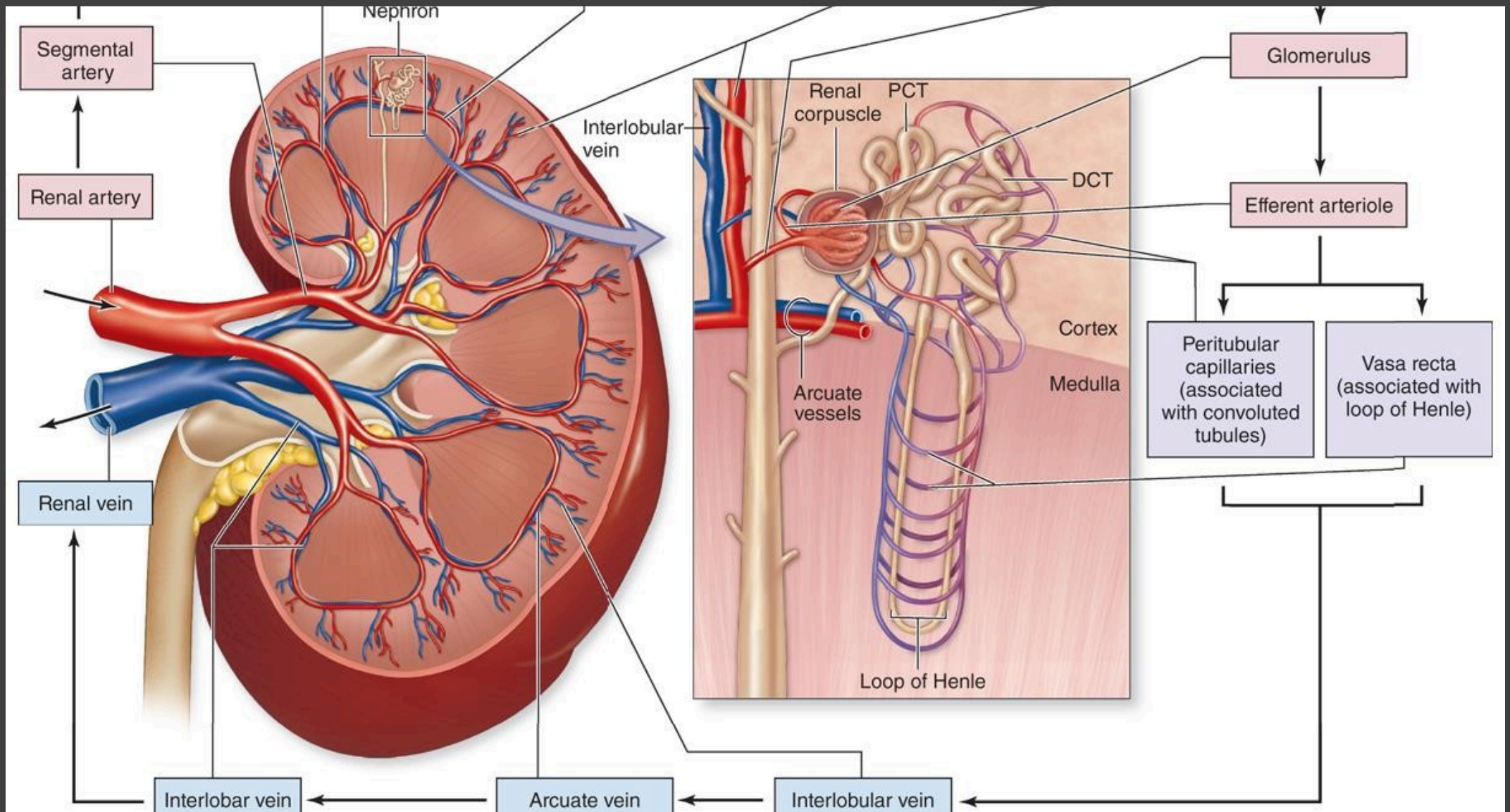
Juxtaglomerular apparatus

Renal interstitial tissue

- -Both the cortex and the medulla contain specialized cells in the spaces between uriniferous tubules and the blood and lymph vessels. These cells are called "interstitial cells".
- -The interstitial cells are more frequent in the medulla where cells containing cytoplasmic lipid droplets are involved in the synthesis of a hypotensor hormone (**medullipin I**) a substance that is converted in the liver to medullipin II, a potent vasodilator that lowers blood pressure.
- -Peritubular capillaries are abundant in the sparse surrounding connective
- tissue interstitium, which fills only about 10% of the cortex .

Blood supply of the kidney:

- As expected for an organ specialized to process the blood, the kidney vasculature is large, well-organized, and closely associated with all components of the nephron. The color seen in the cut surface of the unfixed kidney reflects the distribution of blood in the organ. Approximately 90% to 95% of the blood passing through the kidney is in the cortex; 5% to 10% is in the medulla. Blood vessels of the kidneys are named according to their locations or shapes



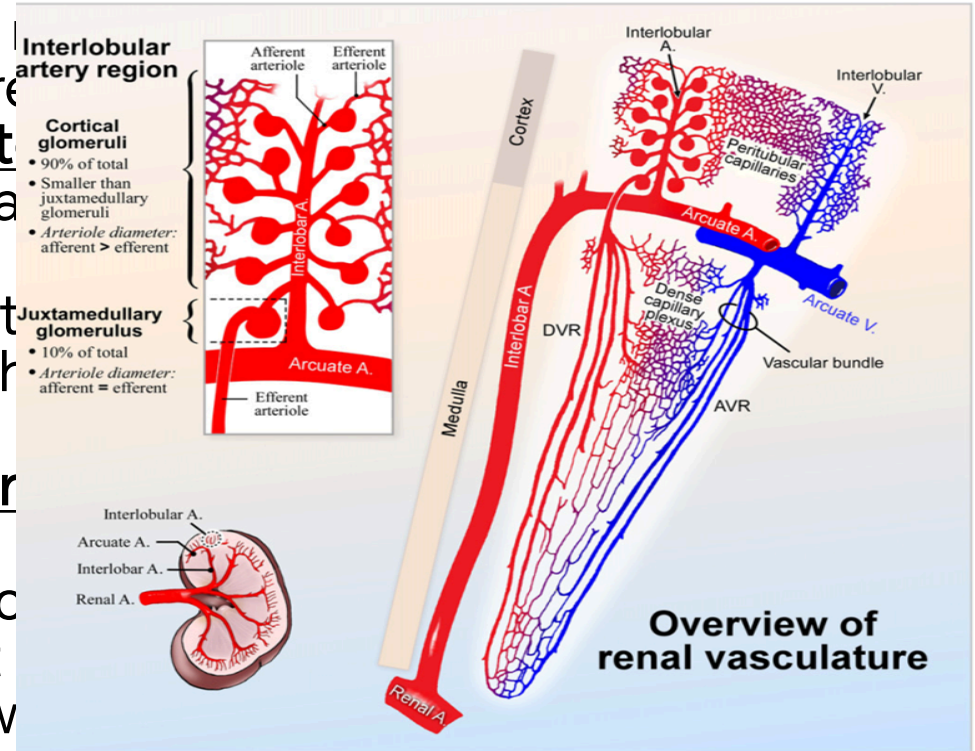
Arterial supply:

Each kidney's renal artery divides into two or three segmental arteries at the hilum. Around the renal pelvis, these arteries branch further as the interlobar arteries, which extend between the renal pyramids toward the corticomedullary junction.

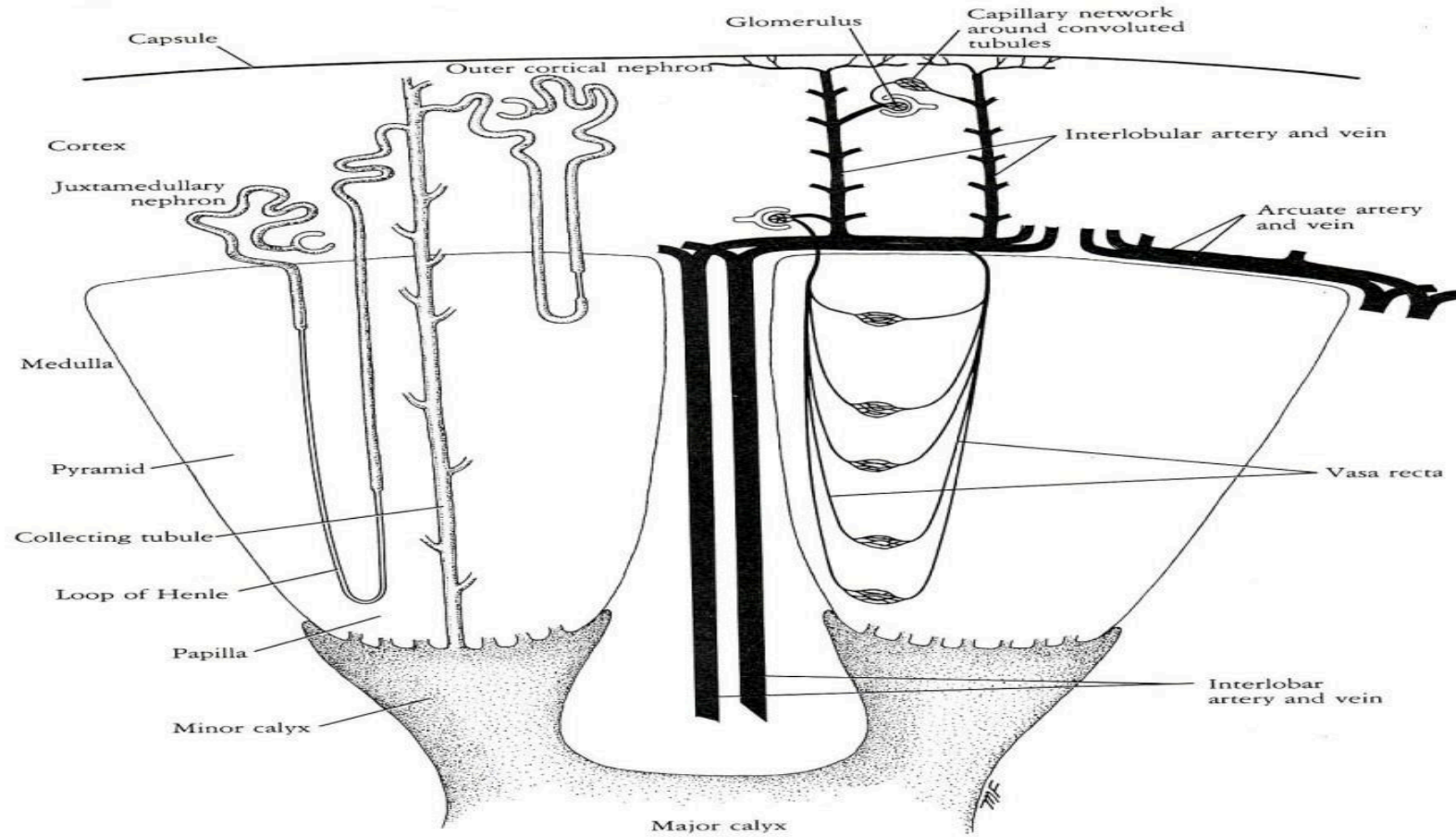
At the junction between the pyramid and cortex, interlobar arteries give arcuate arteries, which then give off interlobular arteries.

The interlobular arteries give off afferent arterioles which form capillary network (glomerulus).

- 4- Blood leaves the glomerular capillaries, not via venules, but via efferent arterioles, which then branch again to form another capillary network usually the peritubular capillaries profusely distributed throughout the cortex.



- 5-From the juxtaglomerular corpuscles near the medulla, **efferent arterioles** do not form peritubular capillaries instead, they branch repeatedly to form parallel tassel-like bundles of capillary loops called the **vasa recta** (L. recta, straight) that penetrate deep into the medulla in association with the loops of Henle and collecting ducts..
- **Collectively, the cortex receives over 10 times** more blood than the medulla. Blood leaves the kidney in veins that follow the same courses as arteries and have the same names.



Blood supply of the kidney

Main kidney functions

1. **Glomerular function**; Excretion of toxic end-products of metabolism and foreign substances.
2. **Tubular function**; Acid-base, water and electrolyte balance.
3. **endocrine activities include**:

- Synthesis and secretion of the glycoprotein hormone erythropoietin (EPO), which acts on the bone marrow and regulates red blood cell formation in response to decreased blood oxygen concentration. EPO is synthesized by **endothelial cells of the peritubular capillaries** in the renal cortex.
- Synthesis and secretion of the acid protease renin, an enzyme involved in control of blood pressure and blood volume. Renin is produced by **juxtaglomerular cells**.
- Hydroxylation of 25-OH vitamin D₃ (a steroid precursor produced in the liver) in the **proximal tubules** of the kidney to produce the highly active 1,25-(OH)₂ vitamin D₃ (calcitriol).



I love Histology!





References:



Junqueira's Basic Histology (Text and Atlas) (2018) 15 th edition.



Kaplan Medical .USMLE® Step 1
Lecture Notes 2020,Anatomy
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Basic_Sciences

QUIZ

Choose the correct answer

- A simple cuboidal epithelial typically lines which of the following?
 - a. A minor calyx.
 - b. The thin part of Henle's loop.
 - c. The proximal convoluted tubule.
 - d. The urethra
- Blood in route to glomeruli flows through which sequence of arteries?
 - a. Interlobar-interlobular-arcuate.
 - b. Renal-interlobar-interlobular.
 - c. Renal-interlobular-arcuate.
 - d. Renal-interlobar-arcuate.